

The Simplex Algorithm

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1 Overview

The simplex algorithm solves linear programs (LPs) of the form

$$\begin{array}{ll}\text{maximize} & c^T x \\ \text{subject to} & Ax \leq b \\ & x \geq 0,\end{array}$$

by moving from one vertex (basic feasible solution) of the feasible polytope to an adjacent vertex, at each step increasing (or not decreasing) the objective, until no adjacent vertex improves it. That final vertex is optimal because the feasible region of a linear program is convex and the objective is linear, so a local optimum among vertices is a global optimum.

2 Geometry of Linear Programming

Polyhedra

The feasible region of an LP,

$$P = \{x \in \mathbb{R}^n : Ax \leq b, x \geq 0\},$$

is a **polyhedron**: an intersection of finitely many half-spaces, one per constraint (including the nonnegativity constraints $x_i \geq 0$). Each half-space is convex, and an intersection of convex sets is convex, so P is always a convex set — it has no dents or holes, and the line segment between any two feasible points is entirely feasible. When P is bounded, it is called a **polytope**: a bounded polyhedron, i.e. the higher-dimensional analogue of a polygon (in $\mathbb{R}^2$) or a polyhedral solid (in $\mathbb{R}^3$).

vertices of P $\longleftrightarrow$ basic feasible solutions of the tableau

Two vertices are **adjacent** — joined by an **edge** of the polyhedron — if their basic feasible solutions differ in exactly one basic variable. This is precisely what a simplex **pivot** does: one variable enters the basis, one leaves, and every other basic variable is recomputed. Geometrically, a pivot is a slide along an edge of the polyhedron from the current vertex to an adjacent one. The entering-variable rule (pick a column with negative reduced cost) is choosing an edge direction that increases the objective; the ratio test is finding exactly how far along that edge you can travel before hitting the *next* constraint boundary — i.e. before you fall off the polyhedron and hit the next vertex.

Why walking vertex to vertex finds the optimum

This relies on the **fundamental theorem of linear programming**: if an LP has an optimal solution, it has one at a vertex of P . Intuitively, a linear objective $c^T x$ has no interior maxima or curvature to get trapped in — its level sets are flat hyperplanes — so if it is not maximized at a vertex, you can always slide along some edge from a non-vertex optimum to strictly improve, or at worst tie, until reaching a vertex. Combined with convexity of P , this means: a vertex where *no adjacent vertex is better* is not just a local optimum among vertices — it is the global optimum over all of P . That is exactly the simplex stopping condition (all z -row entries ≥ 0), so when the algorithm stops, it has certified global optimality, not merely found a good corner.

3 The Simplex Algorithm

Standard form

A constraint $a^T x \leq b$ is converted to an equality by adding a nonnegative *slack variable* s :

$$a^T x + s = b, \quad s \geq 0.$$

A constraint $a^T x \geq b$ uses a *surplus* variable: $a^T x - s = b$. Equality constraints, and $\geq$ constraints after subtracting surplus, generally need an *artificial variable* to supply an obvious starting basic feasible solution; these are handled with the Big-M method (Problems 3-5 below).

The tableau method

Each tableau row corresponds to a basic variable; the bottom row (the objective, or “ z -row”) records reduced costs. One iteration is:

1. **Optimality check.** If every coefficient in the z -row is ≥ 0 (for a max problem written as $z - c^T x = 0$), stop: the current basic feasible solution is optimal.
2. **Entering variable.** Otherwise choose a column with a negative z -row coefficient (commonly the most negative) — this variable enters the basis.
3. **Leaving variable (ratio test).** For each row with a strictly positive entry in the entering column, compute $\frac{\text{RHS}}{\text{entry}}$. The row achieving the minimum ratio determines the variable that leaves the basis. (If all entries in that column are ≤ 0 , the problem is unbounded.)
4. **Pivot.** Use Gauss–Jordan elimination on the pivot entry to make its column a unit column (1 in the pivot row, 0 elsewhere, including the z -row). If y_{rk} is the pivot element (pivot row r , pivot column k), then

$$y'_{rj} = \frac{y_{rj}}{y_{rk}}, \quad y'_{ij} = y_{ij} - \frac{y_{ik} y_{rj}}{y_{rk}} \quad (i \neq r),$$

applied to every column j (including the RHS) and to the objective row as well.

5. Repeat from step 1.

Ties in the ratio test indicate *degeneracy*; ties in the entering-variable choice can be broken arbitrarily (Bland’s rule prevents cycling by always choosing the lowest-indexed eligible variable, if needed).

4 Problem 1 (Introductory): Two Variables

Statement

$$\begin{array}{ll}
 \text{maximize} & z = 3x_1 + 5x_2 \\
 \text{subject to} & x_1 \leq 4 \\
 & 2x_2 \leq 12 \\
 & 3x_1 + 2x_2 \leq 18 \\
 & x_1, x_2 \geq 0
 \end{array}$$

Solution

Add slacks $s_1, s_2, s_3 \geq 0$:

$$x_1 + s_1 = 4, \quad 2x_2 + s_2 = 12, \quad 3x_1 + 2x_2 + s_3 = 18.$$

Initial tableau (basis $\{s_1, s_2, s_3\}$):

	x_1	x_2	s_1	s_2	s_3	RHS
s_1	1	0	1	0	0	4
s_2	0	2	0	1	0	12
s_3	3	2	0	0	1	18
z	-3	-5	0	0	0	0

Reading the z -row

The z -row encodes the equation $z - 3x_1 - 5x_2 = 0$. Its entries are the **reduced costs**: the coefficient under a nonbasic variable is (the negative of) how much z would increase per unit if that variable were increased from 0, holding the current basic variables consistent with the constraints. Three facts about this row are worth tracking through every iteration:

- A basic variable's own column always carries a 0 in the z -row (by construction — that is what “basic” means in this row).
- A negative entry means increasing that variable would *increase* z , since we are maximizing — hence it is a candidate to enter.
- The RHS of the z -row is the objective value of the *current* vertex. Initially all of x_1, x_2 are nonbasic (i.e. = 0), so $z = 0$, matching the vertex $(0, 0)$.

Iteration 1. Most negative z -row entry: x_2 (-5). Ratios: $12/2 = 6$, $18/2 = 9$ (s_1 row has 0, skip). Minimum ratio 6 $\Rightarrow s_2$ leaves, x_2 enters. Pivot on row s_2 , column x_2 (divide row by 2, then eliminate):

	x_1	x_2	s_1	s_2	s_3	RHS
s_1	1	0	1	0	0	4
x_2	0	1	0	1/2	0	6
s_3	3	0	0	-1	1	6
z	-3	0	0	5/2	0	30

How the z -row just changed

Compare this to the initial z -row $[-3, -5, 0, 0, 0 \mid 0]$. The update rule is: new z -row = old z -row $-$ (old z -row entry in the pivot column) $\times$ (new pivot row). Here the pivot column was x_2 (old entry -5) and the new pivot row is $x_2 = [0, 1, 0, 1/2, 0 \mid 6]$, so

$$z\text{-row}_{\text{new}} = [-3, -5, 0, 0, 0 \mid 0] - (-5) \cdot [0, 1, 0, \frac{1}{2}, 0 \mid 6] = [-3, 0, 0, \frac{5}{2}, 0 \mid 30].$$

Three things to notice:

- The x_2 entry became exactly 0 — that column is now basic, as it must be.
- s_2 's entry flipped from 0 to $+5/2$: since s_2 just left the basis (its constraint $2x_2 \leq 12$ is now tight), increasing s_2 again would loosen that binding constraint and *reduce* z — a positive reduced cost for a variable we just vacated is exactly what optimality requires of it.
- The RHS jumped from 0 to 30: this is z at the new vertex $(x_1, x_2) = (0, 6)$, i.e. $3(0) + 5(6) = 30$, matching the geometric picture of having slid from $(0, 0)$ to $(0, 6)$.

x_1 's entry is still $-3 < 0$, so the vertex $(0, 6)$ is not yet optimal — there is still an edge to walk that improves z .

Iteration 2. Only negative entry: x_1 (-3). Ratios: $4/1 = 4$, $6/3 = 2$ (x_2 row has 0, skip). Minimum 2 $\Rightarrow s_3$ leaves, x_1 enters. Pivot on row s_3 , column x_1 :

	x_1	x_2	s_1	s_2	s_3	RHS
s_1	0	0	1	$1/3$	$-1/3$	2
x_2	0	1	0	$1/2$	0	6
x_1	1	0	0	$-1/3$	$1/3$	2
z	0	0	0	$3/2$	1	36

How the z -row changed this time

Same rule again: subtract (old entry in pivot column x_1 , which was -3) times the new pivot row from the old z -row. The new pivot row is $x_1 = [1, 0, 0, -1/3, 1/3 \mid 2]$, so

$$z\text{-row}_{\text{new}} = [-3, 0, 0, \frac{5}{2}, 0 \mid 30] - (-3) \cdot [1, 0, 0, -\frac{1}{3}, \frac{1}{3} \mid 2] = [0, 0, 0, \frac{3}{2}, 1 \mid 36].$$

- x_1 's entry is now 0, since x_1 just became basic.
- s_3 , which just left the basis (its constraint $3x_1 + 2x_2 \leq 18$ is now tight), picked up a positive reduced cost of 1 — again exactly what leaving-the-basis should do to a variable's row here.
- s_2 's reduced cost fell from $5/2$ to $3/2$ but stayed positive, and s_3 's is now 1: both s_2 and s_3 have strictly positive reduced costs, meaning both of their constraints are “expensive” to relax further — consistent with the optimum sitting at the vertex $(2, 6)$ where *both* $2x_2 \leq 12$ and $3x_1 + 2x_2 \leq 18$ are tight.
- $\text{RHS} = 36 = 3(2) + 5(6)$, the objective value at vertex $(2, 6)$ — matching the second red arrow in the picture above.

All z -row entries are now ≥ 0 : no direction out of this vertex improves z , so by the fundamental theorem of linear programming this vertex is not just locally but globally optimal.

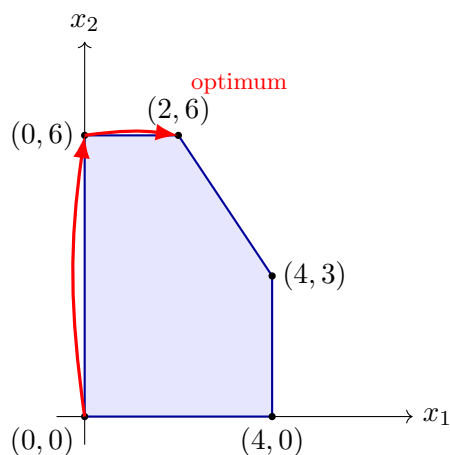
Answer

$$x_1 = 2, \quad x_2 = 6, \quad z^* = 36$$

(This is the classic textbook example; the optimum is the vertex where $2x_2 \leq 12$ and $3x_1 + 2x_2 \leq 18$ are both tight.)

Geometric Picture

Problem 1 is two-dimensional, so its feasible region and simplex path can be drawn directly. The five constraints ($x_1 \leq 4$, $x_2 \leq 6$, $3x_1 + 2x_2 \leq 18$, $x_1 \geq 0$, $x_2 \geq 0$) cut out the pentagon below, with vertices $(0,0)$, $(4,0)$, $(4,3)$, $(2,6)$, $(0,6)$:



The simplex path traced by the two iterations worked out below is exactly the red arrows: start at the origin $(0,0)$ — the BFS with all slacks basic and $x_1 = x_2 = 0$ — slide along an edge to $(0,6)$ (Iteration 1: x_2 enters, s_2 leaves), then along the next edge to $(2,6)$ (Iteration 2: x_1 enters, s_3 leaves), where every remaining direction would decrease z , so the walk stops. Three of the polygon's five vertices are visited; the two vertices $(4,0)$ and $(4,3)$ are never explored because the entering-variable rule always heads toward the steepest available improvement in z , and that path happens not to pass through them.

Problem 2 (Intermediate): Three Variables, Four Constraints

Statement

$$\begin{array}{ll} \text{maximize} & z = 5x_1 + 4x_2 + 3x_3 \\ \text{subject to} & 2x_1 + 3x_2 + x_3 \leq 5 \\ & 4x_1 + x_2 + 2x_3 \leq 11 \\ & 3x_1 + 4x_2 + 2x_3 \leq 8 \\ & x_1, x_2, x_3 \geq 0 \end{array}$$

Solution

Add slacks s_1, s_2, s_3 . Initial tableau:

	x_1	x_2	x_3	s_1	s_2	s_3	RHS
s_1	2	3	1	1	0	0	5
s_2	4	1	2	0	1	0	11
s_3	3	4	2	0	0	1	8
z	-5	-4	-3	0	0	0	0

Iteration 1. Most negative: x_1 (-5). Ratios: $5/2 = 2.5$, $11/4 = 2.75$, $8/3 \approx 2.67$. Minimum $2.5 \Rightarrow s_1$ leaves. Pivot on row s_1 , column x_1 (divide row by 2):

	x_1	x_2	x_3	s_1	s_2	s_3	RHS
x_1	1	$3/2$	$1/2$	$1/2$	0	0	$5/2$
s_2	0	-5	0	-2	1	0	1
s_3	0	$-1/2$	$1/2$	$-3/2$	0	1	$1/2$
z	0	$7/2$	$-1/2$	$5/2$	0	0	$25/2$

Iteration 2. Negative entry: x_3 ($-1/2$). Ratios (positive entries only): row x_1 : $(5/2)/(1/2) = 5$; row s_2 : coefficient 0, skip; row s_3 : $(1/2)/(1/2) = 1$. Minimum 1 $\Rightarrow s_3$ leaves, x_3 enters. Pivot on row s_3 , column x_3 (already coefficient $1/2$, multiply row by 2):

	x_1	x_2	x_3	s_1	s_2	s_3	RHS
x_1	1	$5/2$	0	$5/4$	0	-1	2
s_2	0	-5	0	-2	1	0	1
x_3	0	-1	1	-3	0	2	1
z	0	3	0	1	0	1	13

All z -row entries ≥ 0 : optimal.

Answer

$$\boxed{x_1 = 2, \quad x_2 = 0, \quad x_3 = 1, \quad z^* = 13}$$

Problem 3 (Advanced): A $\geq$ Constraint — Big-M Method

Statement

$$\begin{aligned} &\text{maximize} && z = 2x_1 + 3x_2 \\ &\text{subject to} && x_1 + x_2 \leq 4 \\ &&& x_1 + 2x_2 \geq 4 \\ &&& x_1, x_2 \geq 0 \end{aligned}$$

The second constraint is $\geq$, so the “obvious” slack basis is no longer feasible at the origin; we need the Big-M method.

Solution

Introduce slack $s_1 \geq 0$ on the first constraint, and on the second subtract a surplus $s_2 \geq 0$ and add an artificial variable $a_1 \geq 0$:

$$x_1 + x_2 + s_1 = 4, \quad x_1 + 2x_2 - s_2 + a_1 = 4.$$

Penalize a_1 heavily in the objective: maximize $z = 2x_1 + 3x_2 - Ma_1$, with M a very large constant. Writing $z - 2x_1 - 3x_2 + Ma_1 = 0$ and eliminating a_1 from the z -row (since a_1 is basic in row 2, subtract $M \times \text{row 2}$ from the z -row):

	x_1	x_2	s_1	s_2	a_1	RHS
s_1	1	1	1	0	0	4
a_1	1	2	0	-1	1	4
z	$-2 - M$	$-3 - 2M$	0	M	0	$-4M$

Iteration 1. Most negative z -row coefficient (for large M): x_2 , coefficient $-3 - 2M$. Ratios: $4/1 = 4$ (row s_1), $4/2 = 2$ (row a_1). Minimum 2 $\Rightarrow a_1$ leaves (good — the artificial variable is driven out), x_2 enters. Pivot on row a_1 , column x_2 (divide by 2):

	x_1	x_2	s_1	s_2	a_1	RHS
s_1	$1/2$	0	1	$1/2$	$-1/2$	2
x_2	$1/2$	1	0	$-1/2$	$1/2$	2
z	$-1/2$	0	0	$-3/2$	$M + 3/2$	6

Iteration 2. Negative entry: x_1 ($-1/2$). Ratios: $2/(1/2) = 4$ (row s_1), $2/(1/2) = 4$ (row x_2) — a tie (degenerate). Choose row s_1 to leave. Pivot on row s_1 , column x_1 (multiply by 2):

	x_1	x_2	s_1	s_2	a_1	RHS
x_1	1	0	2	1	-1	4
x_2	0	1	-1	-1	1	0
z	0	0	1	-1	$M + 1$	8

Iteration 3. Negative entry: s_2 (-1). Only row x_1 has a positive entry in that column (1); row x_2 's entry is -1 (skip). Ratio $4/1 = 4 \Rightarrow x_1$ leaves, s_2 enters. Pivot on row x_1 , column s_2 :

	x_1	x_2	s_1	s_2	a_1	RHS
s_2	1	0	2	1	-1	4
x_2	1	1	1	0	0	4
z	1	0	3	0	M	12

All z -row entries ≥ 0 (recall M is huge and positive): optimal. Note $s_2 = 4 \geq 0$ confirms feasibility ($x_1 + 2x_2 = 0 + 8 = 8 \geq 4$, with surplus 4), and $x_1 = 0$.

Answer

$$\boxed{x_1 = 0, \quad x_2 = 4, \quad z^* = 12}$$

Check: $x_1 + x_2 = 4 \leq 4$ (tight), $x_1 + 2x_2 = 8 \geq 4$ (slack), $z = 2(0) + 3(4) = 12$. $\checkmark$

Preliminaries: Max form vs. Min form of the simplex

The simplex algorithm is stated most cleanly for a *maximization* problem in standard form. Every minimization problem can be converted using

$$\min z = -\max(-z), \quad \text{i.e.} \quad z_{\min}^* = -z_{\max}^*, \quad z' := -z.$$

The *argmin* is unchanged; only the optimal value changes sign.

In the Big- M method every artificial variable A_i is given a cost $\pm M$ where $M > 0$ is a symbolic number larger than any number that can arise from the data ($M \rightarrow \infty$). The penalty forces the artificials out of the basis if that is at all possible.

Item	Maximization $\max z = c^\top x$	Minimization $\min z = c^\top x$
Entering variable	most negative $z_j - c_j$	most positive $z_j - c_j$
Stopping criterion	$z_j - c_j \geq 0$ for all j	$z_j - c_j \leq 0$ for all j
Comparing two entries containing M	$\alpha M + \beta$ vs. $\alpha' M + \beta'$: compare α, α' first; use β, β' only to break a tie (since $M \rightarrow \infty$)	identical rule
Leaving variable (min-ratio test)	$\min_{i: y_{ik} > 0} \frac{b_i}{y_{ik}}$	identical (the ratio test never depends on max/min)
Unboundedness	entering column has all $y_{ik} \leq 0$	identical

Problem 4 (Complex): Minimization with Mixed Constraints

Statement

$$\begin{aligned}
 &\text{minimize} && z = 4x_1 + 2x_2 \\
 &\text{subject to} && 3x_1 + x_2 \geq 27 \\
 & && x_1 + x_2 \geq 21 \\
 & && x_1 + 2x_2 \leq 30 \\
 & && x_1, x_2 \geq 0
 \end{aligned}$$

Put $z' := -z$. Then

$$\max z' = -4x_1 - 2x_2, \quad \text{and finally} \quad z_{\min} = -z'_{\max}.$$

4.1 Standard form

Surplus $s_1, s_2 \geq 0$ for the two $\geq$ rows, slack $s_3 \geq 0$ for the $\leq$ row, artificials $A_1, A_2 \geq 0$ for the two $\geq$ rows:

$$\begin{aligned}
 \max \quad & z' = -4x_1 - 2x_2 + 0s_1 + 0s_2 + 0s_3 - MA_1 - MA_2 \\
 \text{s.t.} \quad & 3x_1 + x_2 - s_1 + A_1 = 27 \\
 & x_1 + x_2 - s_2 + A_2 = 21 \\
 & x_1 + 2x_2 + s_3 = 30 \\
 & \text{all variables} \geq 0.
 \end{aligned}$$

Starting basis: $B = \{A_1, A_2, s_3\}$, $x_B = (27, 21, 30)$. We use the $z_j - c_j$ row; **entering** = **most negative** $z_j - c_j$, **stop when all** $z_j - c_j \geq 0$.

4.2 Tableau 1

c_B	B	x_1	x_2	s_1	s_2	s_3	A_1	A_2	b	ratio
$-M$	A_1	3	1	-1	0	0	1	0	27	9 $\leftarrow$
$-M$	A_2	1	1	0	-1	0	0	1	21	21
0	s_3	1	2	0	0	1	0	0	30	30
$z_j - c_j$		$-4M + 4$	$-2M + 2$	M	M	0	0	0	$-48M$	
		$\uparrow$								

$-4M + 4$ is the most negative $\Rightarrow x_1$ enters; ratios $27/3 = 9$, $21/1 = 21$, $30/1 = 30 \Rightarrow A_1$ leaves. Pivot = 3.

4.3 Tableau 2

c_B	B	x_1	x_2	s_1	s_2	s_3	A_1	A_2	b	ratio
-4	x_1	1	$\frac{1}{3}$	$-\frac{1}{3}$	0	0	$\frac{1}{3}$	0	9	27
$-M$	A_2	0	$\frac{2}{3}$	$\frac{1}{3}$	-1	0	$-\frac{1}{3}$	1	12	18
0	s_3	0	$\frac{5}{3}$	$\frac{1}{3}$	0	1	$-\frac{1}{3}$	0	21	$\frac{63}{5} \leftarrow$
$z_j - c_j$		0	$\frac{2}{3} - \frac{2M}{3}$	$\frac{4}{3} - \frac{M}{3}$	M	0	$\frac{4M}{3} - \frac{4}{3}$	0	$-36 - 12M$	
			$\uparrow$							

Compare the M -parts: $-\frac{2}{3}M < -\frac{1}{3}M$, so x_2 enters. Ratios 27, 18, $\frac{63}{5} = 12.6 \Rightarrow s_3$ leaves. Pivot = $\frac{5}{3}$.

4.4 Tableau 3

c_B	B	x_1	x_2	s_1	s_2	s_3	A_1	A_2	b	ratio
-4	x_1	1	0	$-\frac{2}{5}$	0	$-\frac{1}{5}$	$\frac{2}{5}$	0	$\frac{24}{5}$	—
$-M$	A_2	0	0	$\frac{1}{5}$	-1	$-\frac{2}{5}$	$-\frac{1}{5}$	1	$\frac{18}{5}$	18 $\leftarrow$
-2	x_2	0	1	$\frac{1}{5}$	0	$\frac{3}{5}$	$-\frac{1}{5}$	0	$\frac{63}{5}$	63
$z_j - c_j$		0	0	$\frac{6}{5} - \frac{M}{5}$	M	$\frac{2M}{5} - \frac{2}{5}$	$\frac{6M}{5} - \frac{6}{5}$	0	$-\frac{222}{5} - \frac{18M}{5}$	
				$\uparrow$						

Only s_1 has a negative entry ($\frac{6}{5} - \frac{M}{5} < 0$ for large M), so s_1 enters. The x_1 -row entry $-\frac{2}{5} < 0$ is skipped; ratios $\frac{18/5}{1/5} = 18$ and $\frac{63/5}{1/5} = 63 \Rightarrow A_2$ leaves. Pivot = $\frac{1}{5}$.

4.5 Tableau 4 (optimal)

c_B	B	x_1	x_2	s_1	s_2	s_3	A_1	A_2	b
-4	x_1	1	0	0	-2	-1	0	2	12
0	s_1	0	0	1	-5	-2	-1	5	18
-2	x_2	0	1	0	1	1	0	-1	9
$z_j - c_j$		0	0	0	6	2	M	$M - 6$	-66

All $z_j - c_j \geq 0$ (for large M), so the tableau is optimal, and *both artificials have left the basis*, so the solution is feasible for the original problem:

$$\boxed{x_1^* = 12, \quad x_2^* = 9, \quad z'_{\max} = -66 \implies z_{\min} = 66}$$

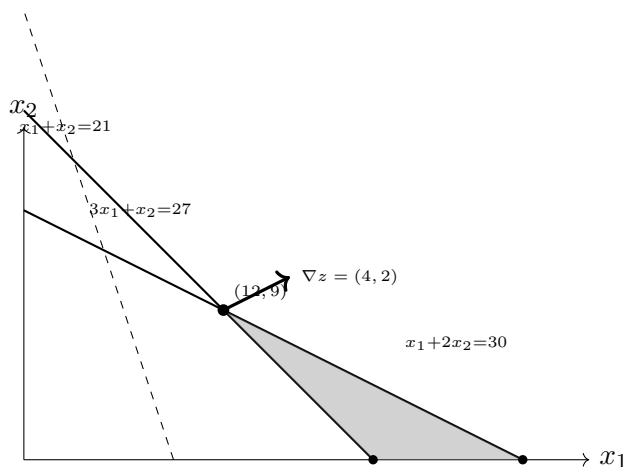
with $s_1 = 18$ (first constraint slack: $3(12) + 9 = 45 \geq 27$), $s_2 = 0$, $s_3 = 0$ (constraints 2 and 3 are *binding*: $12 + 9 = 21$, $12 + 18 = 30$).

4.6 Verification by enumeration of vertices

The feasible region is the triangle with vertices $(21, 0)$, $(30, 0)$, $(12, 9)$:

$$z(21, 0) = 84, \quad z(30, 0) = 120, \quad z(12, 9) = 66.$$

(The intersection $3x_1 + x_2 = 27 \cap x_1 + x_2 = 21$ is $(3, 18)$, which violates $x_1 + 2x_2 \leq 30$; so the first constraint is redundant on the feasible set — consistent with $s_1 = 18 > 0$.)



Problem 5 (Complex): Minimization with Equality and Inequality Constraints

Statement

$$\begin{aligned} &\text{maximize} && z = 5x_1 + 3x_2 \\ &\text{subject to} && x_1 + x_2 = 10 \quad (\text{equality}) \\ & && 2x_1 + x_2 \leq 16 \quad (\leq) \\ & && 3x_1 + x_2 \geq 12 \quad (\geq) \\ & && x_1, x_2 \geq 0. \end{aligned}$$

Because of the equality, the feasible set is a *segment* of the line $x_1 + x_2 = 10$. Substituting $x_2 = 10 - x_1$: the $\leq$ row gives $x_1 \leq 6$, the $\geq$ row gives $x_1 \geq 1$. Hence the feasible set is the segment from $(1, 9)$ to $(6, 4)$, and $z = 30 + 2x_1$ is maximized at $x_1 = 6$. Expected answer: $x^* = (6, 4)$, $z^* = 42$.

4.7 Standard form

Row 1 (equality) needs one artificial A_1 ; row 2 needs a slack s_2 ; row 3 needs a surplus s_3 and an artificial A_2 :

$$\begin{aligned} \max \quad & z = 5x_1 + 3x_2 + 0s_2 + 0s_3 - MA_1 - MA_2 \\ \text{s.t.} \quad & x_1 + x_2 + A_1 = 10 \\ & 2x_1 + x_2 + s_2 = 16 \\ & 3x_1 + x_2 - s_3 + A_2 = 12, \quad \text{all } \geq 0. \end{aligned}$$

4.8 Tableau 1

c_B	B	x_1	x_2	s_2	s_3	A_1	A_2	b	ratio
$-M$	A_1	1	1	0	0	1	0	10	10
0	s_2	2	1	1	0	0	0	16	8
$-M$	A_2	3	1	0	-1	0	1	12	4 $\leftarrow$
$z_j - c_j$		$-4M - 5$	$-2M - 3$	0	M	0	0	$-22M$	
		$\uparrow$							

x_1 enters, A_2 leaves, pivot = 3.

4.9 Tableau 2

c_B	B	x_1	x_2	s_2	s_3	A_1	A_2	b	ratio
$-M$	A_1	0	$\frac{2}{3}$	0	$\frac{1}{3}$	1	$-\frac{1}{3}$	6	9 $\leftarrow$
0	s_2	0	$\frac{1}{3}$	1	$\frac{2}{3}$	0	$-\frac{2}{3}$	8	24
5	x_1	1	$\frac{1}{3}$	0	$-\frac{1}{3}$	0	$\frac{1}{3}$	4	12
$z_j - c_j$		0	$-\frac{2M}{3} - \frac{4}{3}$	0	$-\frac{M}{3} - \frac{5}{3}$	0	$\frac{4M}{3} + \frac{5}{3}$	$20 - 6M$	
			$\uparrow$						

x_2 enters ($-\frac{2}{3}M$ beats $-\frac{1}{3}M$), A_1 leaves, pivot = $\frac{2}{3}$.

4.10 Tableau 3

c_B	B	x_1	x_2	s_2	s_3	A_1	A_2	b	ratio
3	x_2	0	1	0	$\frac{1}{2}$	$\frac{3}{2}$	$-\frac{1}{2}$	9	18
0	s_2	0	0	1	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	5	10 $\leftarrow$
5	x_1	1	0	0	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	1	—
$z_j - c_j$		0	0	0	-1	$M + \frac{5}{2}$	$M - \frac{1}{2}$	32	
					$\uparrow$				

Both artificials are gone, so the current point $(1, 9)$ is feasible ($z = 32$) — it is the *other* endpoint of the segment. Since $z_{s_3} - c_{s_3} = -1 < 0$ we continue: s_3 enters, s_2 leaves, pivot = $\frac{1}{2}$.

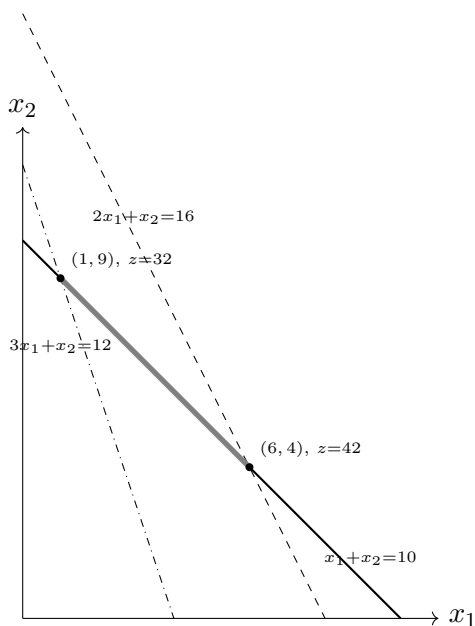
4.11 Tableau 4 (optimal)

c_B	B	x_1	x_2	s_2	s_3	A_1	A_2	b
3	x_2	0	1	-1	0	2	0	4
0	s_3	0	0	2	1	-1	-1	10
5	x_1	1	0	1	0	-1	0	6
$z_j - c_j$		0	0	2	0	$M+1$	M	42

All $z_j - c_j \geq 0$ and no artificial is basic:

$$x_1^* = 6, \quad x_2^* = 4, \quad z^* = 42$$

with $s_2 = 0$ (the $\leq$ row is binding), $s_3 = 10$ (the $\geq$ row is slack: $3(6) + 4 = 22 \geq 12$), and the equality is of course satisfied.



The picture shows the general fact: an equality constraint collapses the feasible set onto a hyperplane (here a line); the $\leq$ and $\geq$ constraints then cut that line down to a segment, whose two endpoints are the only basic feasible solutions.

5 Duality

5.1 The general primal-dual pair

For a maximization primal $\max\{c^\top x : Ax \leq b, x \geq 0\}$ the dual is $\min\{b^\top y : A^\top y \geq c\}$, with the correspondences

Primal (max)	Dual (min)	
i -th constraint $\leq$	dual variable	$y_i \geq 0$
i -th constraint $\geq$	dual variable	$y_i \leq 0$
i -th constraint $=$	dual variable	y_i free
variable $x_j \geq 0$	j -th dual constraint	$\geq c_j$
variable x_j free	j -th dual constraint	$= c_j$
m constraints, n variables	n constraints, m variables	

(For a minimization primal, exchange the two columns.) The dual of the dual is the primal.

5.2 The three theorems

- **Weak duality.** For every primal-feasible x and dual-feasible y , $c^\top x \leq b^\top y$ (max primal). Every dual-feasible point is a *certificate of an upper bound* for the primal.
- **Strong duality.** If either problem has a finite optimum, so does the other and the optimal values are *equal*. Consequently: primal unbounded $\Rightarrow$ dual infeasible, and vice versa.
- **Complementary slackness.** At optimality, for every i and j :

$$y_i (b_i - a_i^\top x) = 0, \quad x_j (a^{(j)\top} y - c_j) = 0.$$

In words: *a non-binding constraint has zero price; a positively priced resource is fully used; a produced good ($x_j > 0$) has its dual constraint tight.*

5.3 Meaning of duality

Economic meaning. If the primal is “choose activity levels x to maximize profit subject to resource availabilities b ”, the dual variable y_i is the *shadow price* (marginal value, imputed price) of one extra unit of resource i :

$$y_i^* = \frac{\partial z^*}{\partial b_i}$$

valid as long as the optimal basis does not change. The dual objective $b^\top y$ is the total imputed value of the endowment, and the dual constraint $\sum_i y_i a_{ij} \geq c_j$ says “the resources consumed by one unit of activity j must be worth at least the profit that activity generates” — an equilibrium/no-arbitrage condition. Duality says the market that prices resources exactly exhausts the firm’s profit.

Algebraic meaning. The dual is a systematic recipe for producing the *best possible bound* obtainable by taking non-negative combinations of the primal constraints. Strong duality asserts that this mechanical bounding procedure is not lossy: the best bound equals the true optimum. This is a special case of the Farkas lemma / theorems of the alternative.

Computational meaning. The simplex method runs the primal and dual simultaneously: at every iteration the basis gives a primal-feasible $x_B = B^{-1}b$ and a dual vector $y^\top = c_B^\top B^{-1}$ which satisfies complementary slackness automatically. The entries of the objective row are exactly the dual slacks $z_j - c_j = y^\top a^{(j)} - c_j$. The stopping criterion $z_j - c_j \geq 0$ is precisely “ y has become

dual-feasible". Hence: *simplex terminates when primal feasibility and dual feasibility hold together*, which by strong duality is optimality. This is also why the dual solution can be read straight off the final tableau, and why the dual simplex method (keep dual feasibility, restore primal feasibility) exists.

5.4 Dual of Problem 4

Primal (a minimization). Write the $\leq$ row as $-x_1 - 2x_2 \geq -30$ so all rows are $\geq$; attach $y_1, y_2, y_3 \geq 0$ to the three rows:

$$\begin{aligned} \text{maximize} \quad & w = 27y_1 + 21y_2 - 30y_3 \\ \text{s.t.} \quad & 3y_1 + y_2 - y_3 \leq 4 \quad (\text{from } x_1) \\ & y_1 + y_2 - 2y_3 \leq 2 \quad (\text{from } x_2) \\ & y_1, y_2, y_3 \geq 0. \end{aligned}$$

Reading the dual from the final tableau. In Tableau 4 the objective-row entries of the slack/surplus columns are

$$s_1 : 0, \quad s_2 : 6, \quad s_3 : 2 \implies y^* = (y_1, y_2, y_3) = (0, 6, 2),$$

and $w^* = 27(0) + 21(6) - 30(2) = 126 - 60 = 66 = z_{\min}$. **Strong duality verified.**

Complementary slackness check.

- $3x_1 + x_2 = 45 > 27$ (slack) $\Rightarrow y_1 = 0$. ✓
- $y_2 = 6 > 0 \Rightarrow x_1 + x_2 = 21$ binding. ✓
- $y_3 = 2 > 0 \Rightarrow x_1 + 2x_2 = 30$ binding. ✓
- $x_1 = 12 > 0 \Rightarrow$ first dual constraint tight: $3(0) + 6 - 2 = 4$. ✓
- $x_2 = 9 > 0 \Rightarrow$ second dual constraint tight: $0 + 6 - 2(2) = 2$. ✓

Interpretation. Think of x_1, x_2 as amounts of two inputs bought at unit costs 4 and 2, which must meet requirements 27 and 21 while respecting a capacity 30. Then $y_2 = 6$ says: *raising the requirement $x_1 + x_2 \geq 21$ to 22 raises the minimum cost by 6* (indeed the new optimum is $(14, 8)$ with $z = 72$). And $y_3 = 2$ attached to the $\leq$ row says: *relaxing the capacity from 30 to 31 lowers cost by 2* (new optimum $(11, 10)$, $z = 64$). The requirement $3x_1 + x_2 \geq 27$ is free of charge, $y_1 = 0$: it is not restricting the optimum at all.

Geometry for Problem 4. At $x^* = (12, 9)$ the active constraints are $x_1 + x_2 \geq 21$, with inward normal $n_2 = (1, 1)$, and $x_1 + 2x_2 \leq 30$, with inward normal $n_3 = (-1, -2)$. The objective gradient is $c = (4, 2)$, and indeed

$$c = (4, 2) = 6(1, 1) + 2(-1, -2) = y_2 n_2 + y_3 n_3,$$

with $y_2 = 6 \geq 0$, $y_3 = 2 \geq 0$: the cost vector sits inside the cone spanned by the two active inward normals, which is exactly the geometric certificate of optimality. The inactive constraint's normal $(3, 1)$ receives weight $y_1 = 0$.

5.5 Dual of Problem 5 (mixed constraints)

Primal is a maximization with an $=$, a $\leq$ and a $\geq$ row, so y_1 free, $y_2 \geq 0$, $y_3 \leq 0$:

$$\begin{aligned} \text{minimize} \quad & w = 10y_1 + 16y_2 + 12y_3 \\ \text{s.t.} \quad & y_1 + 2y_2 + 3y_3 \geq 5 \\ & y_1 + y_2 + y_3 \geq 3 \\ & y_1 \text{ free, } y_2 \geq 0, y_3 \leq 0. \end{aligned}$$

From the final tableau: s_2 -column entry $= 2 \Rightarrow y_2 = 2$; s_3 is basic $\Rightarrow y_3 = 0$; the A_1 -column entry is $M + 1$, so stripping the penalty, $y_1 = 1$. Hence $y^* = (1, 2, 0)$ and

$$w^* = 10(1) + 16(2) + 12(0) = 42 = z^*. \quad \checkmark$$

Both dual constraints are tight ($1 + 4 + 0 = 5$, $1 + 2 + 0 = 3$), consistent with $x_1, x_2 > 0$; and $y_3 = 0$ matches the slack $\geq$ constraint ($22 > 12$). Note how the three constraint types show up: the free variable y_1 (equality row) is allowed to take either sign, the $\leq$ row has $y_2 = 2 > 0$ because it is binding, and the slack $\geq$ row is priced at 0.

Geometry for Problem 5. Here $c = (5, 3)$ and the active constraints at $(6, 4)$ are the equality $x_1 + x_2 = 10$ (normal $(1, 1)$, sign unrestricted because an equality can push both ways) and $2x_1 + x_2 \leq 16$ (outward normal $(2, 1)$):

$$(5, 3) = 1 \cdot (1, 1) + 2 \cdot (2, 1),$$

i.e. $y_1 = 1$, $y_2 = 2$ — again exactly the dual solution. Since the equality's multiplier is unrestricted in sign, the “cone” is really a cone plus a line, which is the geometric counterpart of y_1 being a free variable.

	Problem 4	Problem 5
Primal optimum	$x^* = (12, 9)$	$x^* = (6, 4)$
Optimal value	$z_{\min} = 66$ ($z'_{\max} = -66$)	$z_{\max} = 42$
Artificials used	A_1, A_2 (two $\geq$ rows)	A_1 ($=$ row), A_2 ($\geq$ row)
Pivots	3	3
Binding rows	rows 2 and 3	rows 1 and 2
Dual optimum	$y^* = (0, 6, 2)$, $w^* = 66$	$y^* = (1, 2, 0)$, $w^* = 42$